

Abdulaziz Alhaidan

AI / Machine Learning Engineer

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SUMMARY

AI / Machine Learning Engineer with a B.Sc. in Computer Science who designs, builds, and ships production-grade LLM and ML systems end-to-end. Won 1st place at the Himmah Digital Camps hackathon for a multi-agent executive-advisory system, and deployed an edge-optimized computer-vision model at 73 ms inference latency. Proficient in Python across the full lifecycle — data preparation, model training, evaluation, and production deployment via REST APIs. Specialized in LLMs, multi-agent orchestration, RAG pipelines, embeddings, and vector databases (FAISS, ChromaDB), with hands-on integration of OpenAI and Anthropic Claude into reliable, maintainable services.

TECHNICAL SKILLS

Languages: Python (primary), JavaScript, Java, C++

Generative AI & LLMs: LLM integration (OpenAI, Anthropic Claude), multi-agent systems & agent orchestration, RAG (built from scratch), embeddings, vector databases (FAISS, ChromaDB), indexing; Transformers & BERT, Hugging Face, fine-tuning, attention mechanism; Prompt Engineering (prompt chaining, DSPy/GEPA, structured extraction)

ML Frameworks: PyTorch, TensorFlow, Scikit-learn

Deep Learning: CNNs (transfer learning, ResNet50; CIFAR-10, Fashion MNIST), RNN/LSTM, Seq2Seq, time-series forecasting; PyTorch fundamentals (tensors, autograd, custom NNs)

Classical ML: Linear/Logistic Regression, Decision Trees, Random Forest, Ensembles (Bagging, Voting), K-Means, KNN; evaluation & diagnostics (confusion matrices, precision/recall, failure-case analysis)

Data & MLOps: EDA (pandas, PyGWalker), feature engineering, hyperparameter tuning (GridSearchCV), MLflow experiment tracking, Conda environments

Model Packaging & Deployment: Serialization (Pickle, Joblib), ONNX export (Scikit-learn / PyTorch), FastAPI model serving, edge deployment (Raspberry Pi)

Backend & APIs: FastAPI, Flask, REST APIs, API integration

Cloud & DevOps: AWS (IAM, EC2, S3), Docker, CI/CD fundamentals, Railway

Databases: SQL, MySQL, vector databases, database design

Frontend: React.js, HTML, CSS

Practices: Responsible AI / AI ethics, Git, version control

SELECTED PROJECTS

AI Board Room — Multi-Agent Executive Advisory System — 1st Place, Himmah Digital Camps

- Built a multi-agent system that simulates a corporate advisory board, with specialized agents (CFO, Legal Advisor, Market Strategist) that deliberate, vote, and produce consensus recommendations synthesized by a Chairman agent.
- Engineered a RAG layer over company documents with FAISS so each agent grounds its reasoning in retrieved, source-specific context.
- Orchestrated multi-agent workflows and tool calls with LangChain; served the system through a FastAPI backend with a React interface.

Tech: Python, LangChain, FAISS, FastAPI, React

AgroCure — Plant Disease Detection & Advisory System

- Built an end-to-end plant-disease classification system (ResNet50 transfer learning, 36+ classes) on ~7,300 images using 5-fold cross-validation and a WeightedRandomSampler to address class imbalance.
- Added a retrieval (RAG) layer for rare classes and fine-tuned a new class (Lemon_Healthy) on a custom 80-image dataset collected and labeled in-house.
- Deployed to a Raspberry Pi for on-device inference at 73 ms latency; tracked experiments and model selection with MLflow on DagsHub.

- Led a four-person engineering team across data pipeline, training, and deployment.

Tech: Python, PyTorch, ResNet50, FAISS, MLflow

Aime — Arabic-First AI Voice Assistant

- Production-deployed Arabic-first voice assistant integrating OpenAI Whisper (STT), Anthropic Claude (reasoning), and ElevenLabs (TTS, voice cloning).
- Implemented a RAG pipeline with LangChain + ChromaDB for grounded, context-aware responses.
- Served via a FastAPI backend and deployed as a web app.

Tech: Python, FastAPI, LangChain, ChromaDB, OpenAI Whisper, Anthropic Claude

Between the Lines — AI X-Archive Analyzer

- AI-powered X (Twitter) archive analyzer that surfaces hidden engagement patterns and generates actionable content insights using the Claude API.
- Built client-side ZIP parsing for fully in-browser processing, plus interactive dashboards with heatmaps, trend charts, and weekly content plans.

Tech: Next.js, FastAPI, Claude API

Additional Projects

- **Weather-Driven Car Wash Demand Forecasting** — ML system predicting demand from external weather APIs; forecasting models in Scikit-learn served through a FastAPI REST API.
- **Car Wash Management Platform** — Full-stack booking, order-tracking, and payment system with role-based REST APIs for mobile and admin apps; optimized database queries for performance.
- **Online Multiplayer Party Game** — Real-time multiplayer architecture with React and Socket.IO, including concurrent state synchronization across users.

AI / Machine Learning Engineering — Bootcamp Projects & Coursework

2026 – Present

- Designed end-to-end ML pipelines from EDA to evaluation, applying feature engineering and hyperparameter tuning (GridSearchCV).
- Built classification and regression models with rigorous error analysis (confusion matrices, failure-case inspection); applied clustering (K-Means, KNN) for anomaly detection and segmentation.
- Trained CNNs (CIFAR-10, Fashion MNIST) and sequence models (RNN/LSTM, Seq2Seq); built time-series forecasting models.
- Fine-tuned transformer models with Hugging Face (BERT for sentiment analysis, text classification, word similarity).
- Built a complete RAG system from scratch with embeddings and a FAISS vector database; applied advanced prompt engineering (prompt chaining, DSPy/GEPA) for structured extraction.
- Prepared models for production via serialization (Pickle/Joblib) and ONNX export; tracked experiments with MLflow inside organized Conda environments.

PROFESSIONAL EXPERIENCE

Quality Assurance Specialist — T2, Riyadh, Saudi Arabia

Feb 2025 – May 2025

- Executed functional and regression testing, reported and tracked defects, and validated fixes across releases to ensure product quality.

EDUCATION

B.Sc. in Computer Science — Majmaah University · GPA 4.01 / 5

2020 – 2025

LICENSES & CERTIFICATIONS

- IBM Data Science Professional Certificate
- DeepLearning.AI TensorFlow Specialization
- AWS Cloud Practitioner Essentials

LANGUAGES

Arabic (Native) · English (Advanced)